

# Multi-Subsystem Quantum Non-Demolition Classification: Complete Commutation, Single-Particle Thermodynamics, and Simultaneous Detection at the LHC

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## Abstract

We prove that the four primary LHC detector subsystems — inner tracker, electromagnetic calorimeter, hadronic calorimeter, and muon spectrometer — measure four mutually commuting observables on the partition coordinate space  $(n, \ell, m, s)$ . The complete commutation relation  $[\hat{O}_n, \hat{O}_\ell] = [\hat{O}_\ell, \hat{O}_m] = [\hat{O}_m, \hat{O}_s] = \dots = 0$  (Theorem 2.3) establishes these as a Complete Set of Commuting Observables (CSCO), and the Quantum Non-Demolition (QND) property (Theorem 3.2) shows that measuring any one coordinate does not disturb the others. The fractional momentum backaction is bounded by  $\Delta p/p \leq \lambda_{\text{dB}}/(2\pi L) \approx 4 \times 10^{-16}$  for typical LHC conditions (Theorem 3.4) — fourteen orders of magnitude below the detector’s intrinsic momentum resolution.

We resolve a foundational paradox of single-particle thermodynamics: classical thermodynamics requires  $N \rightarrow \infty$  for temperature and pressure to be defined, yet every detected LHC event involves a finite, often small, number of particles. The resolution is the *categorical temperature*  $T_{\text{cat}} = \hbar\omega_c/(2\pi k_B)$  (Definition 4.1), which enables the *single-particle ideal gas law*  $PV = k_B T_{\text{cat}}$  for  $N = 1$  (Theorem 4.4). For the LHC trigger reference oscillator at 40 MHz,  $T_{\text{cat}} \approx 0.31$  nK; for a pion with  $p_T = 1$  GeV/c in  $B = 2$  T,  $T_{\text{cat}} \approx 5.3$   $\mu$ K.

We prove the *Fundamental Identity*  $dM/dt = \omega/(2\pi) = 1/\langle\tau_p\rangle$  (Theorem 5.1) unifying temporal evolution, oscillatory dynamics, and categorical state counting as three descriptions of one process. We prove *heat-entropy decoupling*  $\text{Cov}(\delta Q, dS_{\text{cat}}) = 0$  (Theorem 5.4): detector thermal noise and physics signal are statistically independent because they occupy orthogonal degrees of freedom. This is the formal reason timing-only triggers can discriminate signal from background using  $\Delta P(k)$  alone.

We introduce the *Kuramoto order parameter* for the LHC detector  $R_c = \exp(-2\pi^2 \cdot \text{CV}^2)$  where  $\text{CV} = \text{RMSSD}/\bar{T}_{BX}$ , and prove that the five operational detector regimes (turbulent  $R_c < 0.3$ , aperture 0.3–0.5, cascade 0.5–0.8, coherent 0.8–0.95, phase-locked  $R_c > 0.95$ ) correspond to the five thermodynamic regimes of de-

tected ions (Theorem 6.3), establishing a precise mapping from detector synchronisation quality to event classification quality.

**Keywords:** quantum non-demolition; complete commuting observables; single-particle thermodynamics; categorical temperature; fundamental identity; heat-entropy decoupling; Kuramoto order parameter; detector regimes; LHC trigger; partition coordinates

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## 1 Introduction

### 1.1 The multi-subsystem problem

The ATLAS detector at the LHC comprises four primary detection subsystems, each exploiting a different physical mechanism: the inner tracker measures charged-particle momenta via curvature in a 2 T solenoid field; the electromagnetic calorimeter measures electron and photon energies via shower development in liquid argon; the hadronic calorimeter measures hadron energies via shower development in steel/scintillator; and the muon spectrometer identifies muons via track reconstruction in air-core toroids (ATLAS Collaboration, 2008).

These four subsystems produce four independent measurements of each traversing particle. The trigger must combine all four simultaneously in the L1 decision — within a fixed latency of 100 bunch crossings.

The question this paper addresses: *is simultaneous combination of all four subsystems fundamentally consistent with quantum mechanics?* The answer, which is not obvious a priori, is yes — but only because the four measured observables belong to the partition coordinate system  $(n, \ell, m, s)$  of bounded phase space, which has the special property of complete mutual commutation. For conventional quantum observables (position, momentum), simultaneous measurement is forbidden by the Heisenberg uncertainty principle. For partition coordinates, simultaneous measurement is guaranteed by the CSCO structure.

### 1.2 Overview of results

The paper derives and proves:

- (1) Complete commutation of all four detector observables, enabling simultaneous measurement without ordering constraints (Theorem 2.3).
- (2) QND property: measuring any partition coordinate preserves all others (Theorem 3.2).
- (3) Backaction bound:  $\Delta p/p \approx 4 \times 10^{-16}$ , negligible at any LHC energy (Theorem 3.4).
- (4) Resolution-time tradeoff:  $R = m/\Delta m = \omega\tau/(2\pi)$  — resolution scales with integration time (Theorem 3.7).

- (5) Single-particle ideal gas:  $PV = k_B T_{\text{cat}}$  for  $N = 1$  (Theorem 4.4).
- (6) Categorical temperature values for LHC particles (Section 4.4).
- (7) Fundamental identity  $dM/dt = 1/\langle\tau_p\rangle$  (Theorem 5.1).
- (8) Heat-entropy decoupling: thermal noise and physics signal are statistically independent (Theorem 5.4).
- (9) Counting irreversibility: trigger decisions cannot be undone (Theorem 5.7).
- (10) Five detector operational regimes from the Kuramoto order parameter (Theorem 6.3).

## 2 Partition Coordinates as Quantum Observables

### 2.1 Coordinate definitions and detector realizations

From the partition algebra of bounded phase space (Sachikonye, b), every particle in a bounded three-dimensional oscillatory system carries partition coordinates  $(n, \ell, m, s)$ :

- $n \geq 1$ : principal level (radial closure number).
- $0 \leq \ell \leq n - 1$ : angular sublevel (polar closure).
- $-\ell \leq m \leq +\ell$ : azimuthal projection.
- $s \in \{-\frac{1}{2}, +\frac{1}{2}\}$ : rotational parity.

At the LHC, each of these is measured by a dedicated detector subsystem:

**Definition 2.1** (LHC Partition Coordinate Operators). The four partition coordinate operators for a particle traversing the ATLAS detector are:

- $\hat{O}_n = f(\hat{H})$ : principal number, measured by the inner tracker as the momentum scale  $n = \lfloor \sqrt{p_T/p_{\text{ref}}} \rfloor + 1$  where  $p_{\text{ref}} = 1 \text{ GeV}/c$ .
- $\hat{O}_\ell = g(\hat{L}^2)$ : angular complexity, measured by the EM calorimeter as the shower shape parameter —  $\ell = 0$  for electromagnetic (narrow, symmetric),  $\ell \geq 1$  for hadronic (wide, structured).
- $\hat{O}_m = h(\hat{L}_z)$ : orientation, measured by the hadronic calorimeter as the energy partition between EM and hadronic components, encoding the azimuthal angular momentum projection.
- $\hat{O}_s = k(\hat{S}_z)$ : chirality, measured by the muon spectrometer as the track curvature sign —  $s = +\frac{1}{2}$  for positive charge,  $s = -\frac{1}{2}$  for negative.

**Remark 2.2** (HL-LHC fifth coordinate). The HL-LHC precision timing detector (HGTD, CMS ETL) will measure a fifth coordinate — the phase of particle arrival relative to the bunch clock at 20–30 ps precision. This provides an independent handle on the parity/temporal coordinate, extending the system to  $d = 4$  independent subsystems and reducing the Planck depth from  $n_P = 60$  to 52 (Sachikonye, c).

### 2.2 Complete commutation

**Theorem 2.3** (Complete Commutation of Detector Observables). *The four detector observables of Defini-*

tion 2.1 satisfy complete mutual commutation:

$$[\hat{O}_n, \hat{O}_\ell] = [\hat{O}_\ell, \hat{O}_m] = [\hat{O}_m, \hat{O}_s] = [\hat{O}_n, \hat{O}_m] = [\hat{O}_n, \hat{O}_s] = [\hat{O}_\ell, \hat{O}_s] = 0 \quad (1)$$

They form a Complete Set of Commuting Observables (CSCO) on the partition coordinate space.

*Proof.* Each operator is a function of a conserved quantity of the particle dynamics in the detector:

- $\hat{O}_n = f(\hat{H})$  (energy):  $[\hat{H}, \hat{H}] = 0$ , so  $[\hat{O}_n, \hat{H}] = 0$ . Since  $\hat{O}_n$  depends only on  $\hat{H}$ , it commutes with all other conserved quantities.
- $\hat{O}_\ell = g(\hat{L}^2)$ : for the approximately spherically symmetric EM shower,  $[\hat{L}^2, \hat{H}] = 0$  and  $[\hat{L}^2, \hat{L}_z] = 0$  (standard angular momentum algebra).
- $\hat{O}_m = h(\hat{L}_z)$ : for  $z$ -axis-aligned tracking,  $[\hat{L}_z, \hat{H}] = 0$  and  $[\hat{L}_z, \hat{L}^2] = 0$ .
- $\hat{O}_s = k(\hat{S}_z)$ : electric charge is an internal quantum number,  $[\hat{S}_z, \hat{H}] = [\hat{S}_z, \hat{L}^2] = [\hat{S}_z, \hat{L}_z] = 0$ .

All four are functions of the mutually commuting set  $\{\hat{H}, \hat{L}^2, \hat{L}_z, \hat{S}_z\}$ , so any two functions of elements from this set commute.

The CSCO property follows: the four operators simultaneously diagonalize on a common eigenbasis (the partition states), and their eigenvalues  $(n, \ell, m, s)$  uniquely specify each state.  $\square$

**Corollary 2.4** (No Uncertainty Principle for Partition Coordinates). *All four partition coordinates can be measured simultaneously with arbitrary precision:  $\Delta n \cdot \Delta \ell \cdot \Delta m \cdot \Delta s = 0$ . No generalised uncertainty relation constrains joint measurement.*

*Proof.* The Robertson-Schrödinger uncertainty relation states  $\Delta A \cdot \Delta B \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|$ . By Theorem 2.3, all commutators vanish, so the lower bound is zero. Simultaneous eigenstates with sharp values for all four coordinates exist and are precisely the partition states.  $\square$

*Remark 2.5* (Contrast with position-momentum). The uncertainty principle  $\Delta x \cdot \Delta p \geq \hbar/2$  arises because  $[\hat{x}, \hat{p}] = i\hbar \neq 0$ . The partition coordinates commute because they label discrete equivalence classes of trajectories, not continuously varying dynamical variables. This is the crucial distinction: partition coordinates are categorical, not dynamical. They describe *which class* a trajectory belongs to, not *where* in phase space the particle is.

## 3 Quantum Non-Demolition Measurement

### 3.1 The QND criterion

**Definition 3.1** (QND Observable). An observable  $\hat{O}$  is *quantum non-demolition* (QND) if it commutes with the Hamiltonian:  $[\hat{O}, \hat{H}] = 0$ . Equivalently,  $\hat{O}$  is a constant of motion:  $d\hat{O}/dt = (i/\hbar)[\hat{H}, \hat{O}] = 0$ . A QND measurement leaves the post-measurement state as an eigenstate of  $\hat{O}$ , so repeated measurements yield the same result without disturbing the state.

**Theorem 3.2** (QND Property of Partition Coordinates). *Each partition coordinate operator  $\hat{O}_n, \hat{O}_\ell, \hat{O}_m, \hat{O}_s$  satisfies the QND condition  $[\hat{O}_i, \hat{H}] = 0$ . The detector measurement of partition coordinates is QND: repeated measurements yield identical results, and measuring any coordinate does not disturb the others.*

*Proof.* From the proof of Theorem 2.3, each operator is a function of a conserved quantity  $(\hat{H}, \hat{L}^2, \hat{L}_z, \hat{S}_z)$ , all of which commute with  $\hat{H}$  for the relevant symmetries of the detector:

- $[\hat{O}_n, \hat{H}] = [f(\hat{H}), \hat{H}] = 0$  (trivially).
- $[\hat{O}_\ell, \hat{H}] = [g(\hat{L}^2), \hat{H}] = 0$  for central/approximately central potentials.
- $[\hat{O}_m, \hat{H}] = [h(\hat{L}_z), \hat{H}] = 0$  for  $z$ -axis symmetry.
- $[\hat{O}_s, \hat{H}] = 0$  since charge is a conserved quantum number.

The Heisenberg equation  $d\hat{O}/dt = (i/\hbar)[\hat{H}, \hat{O}] = 0$  makes each a constant of motion.

QND repeated measurement: at time  $t_1$ , the system is prepared in eigenstate  $|n, \ell, m, s\rangle$ . By the Schrödinger equation, it evolves as  $|n, \ell, m, s, t\rangle = e^{-i\hat{H}t/\hbar}|n, \ell, m, s\rangle = e^{-iE_{n\ell m s}t/\hbar}|n, \ell, m, s\rangle$  (still an eigenstate with the same quantum numbers). Measurement at  $t_2 > t_1$  finds the same values  $(n, \ell, m, s)$ .

Non-disturbance: since all four operators commute (Theorem 2.3), they share a common eigenbasis. Measuring  $\hat{O}_n$  projects onto a subspace of constant  $n$ , which is also an eigenspace of  $\hat{O}_\ell, \hat{O}_m$ , and  $\hat{O}_s$ . No collapse of the other coordinates occurs.  $\square$

**Corollary 3.3** (Non-Destructive Particle Identification). *The trigger can determine  $(n, \ell, m, s)$  of each detected particle from a single measurement event (one detector crossing), with all four coordinates sharp and all subsequent measurements consistent. The identification is non-destructive: the particle's state after classification is the same as before.*

### 3.2 Measurement backaction bound

**Theorem 3.4** (Backaction Suppression). *The fractional momentum disturbance from partition coordinate measurement is bounded by:*

$$\frac{\Delta p}{p} \leq \frac{\hbar}{L \cdot p} = \frac{\lambda_{\text{dB}}}{2\pi L}, \quad (2)$$

where  $\lambda_{\text{dB}} = h/p$  is the de Broglie wavelength and  $L$  is the characteristic detector cell size.

*Proof.* Categorical measurement localises the particle to a phase-space cell of linear size  $\delta x \sim L/n$  (at partition level  $n$ ). By the Heisenberg uncertainty principle, the minimum momentum uncertainty induced is:

$$\Delta p \geq \frac{\hbar}{2\delta x} = \frac{n\hbar}{2L} \geq \frac{\hbar}{2L}$$

(the last inequality uses  $n \geq 1$ ). The fractional disturbance is:

$$\frac{\Delta p}{p} = \frac{\hbar}{2Lp} = \frac{h}{4\pi Lp} = \frac{\lambda_{\text{dB}}}{4\pi L} \leq \frac{\lambda_{\text{dB}}}{2\pi L}.$$

$\square$

**Example 3.5** (Backaction at the LHC). For a pion with  $p_T = 1 \text{ GeV}/c$  ( $\lambda_{\text{dB}} \approx 1.2 \times 10^{-15} \text{ m}$ ) and tracker cell size  $L = 50 \mu\text{m} = 5 \times 10^{-5} \text{ m}$ :

$$\frac{\Delta p}{p} \approx \frac{1.2 \times 10^{-15}}{2\pi \times 5 \times 10^{-5}} \approx 3.8 \times 10^{-12}.$$

For a 10 GeV pion ( $10\times$  higher momentum,  $10\times$  smaller  $\lambda_{\text{dB}}$ ):  $\Delta p/p \approx 3.8 \times 10^{-13}$ . These are 10–11 orders of magnitude below the ATLAS tracker’s intrinsic momentum resolution of  $\sigma(p)/p \sim 10^{-2}$ . The quantum backaction is entirely negligible at any foreseeable LHC energy.

*Remark 3.6* (Classical interpretation). At LHC energies, the de Broglie wavelength of all detected particles is many orders of magnitude smaller than any detector cell. The system operates in the deep classical limit where  $\lambda_{\text{dB}}/L \ll 1$  and the backaction vanishes. The QND analysis is therefore not a correction to classical detector physics — it is a confirmation that classical detector operation is in perfect agreement with quantum mechanics.

### 3.3 Resolution-time tradeoff

**Theorem 3.7** (Resolution-Time Tradeoff). *For a particle executing  $M$  categorical state transitions per oscillation cycle at frequency  $\omega$ , observed for time  $\tau$ , the mass (momentum) resolution is:*

$$R = \frac{p}{\Delta p} = \frac{\omega\tau}{2\pi}. \quad (3)$$

*Resolution scales linearly with observation time.*

*Proof.* The number of complete oscillation cycles in time  $\tau$  is  $N_{\text{cycles}} = \omega\tau/(2\pi)$ . Each cycle contributes one independent partition state measurement. The momentum resolution from  $N_{\text{cycles}}$  independent measurements improves as  $\sqrt{N_{\text{cycles}}}$  (for independent Gaussian errors), but for the partition framework each cycle counts one state exactly (integer counting, not continuous signal), so the resolution equals the number of resolved states:  $R = N_{\text{cycles}} = \omega\tau/(2\pi)$ .  $\square$

**Example 3.8** (LHC inner tracker resolution). For a silicon tracker with 100 MHz readout frequency ( $\omega = 2\pi \times 10^8 \text{ rad/s}$ ) and integration time  $\tau = 25 \text{ ns}$  (one bunch crossing):  $R = 10^8 \times 25 \times 10^{-9} = 2.5$  — only 2.5 independent partition states per bunch crossing per tracker layer.

For a drift tube muon chamber with 1 MHz drift frequency and 50 ns integration:  $R = 10^6 \times 50 \times 10^{-9} = 0.05$  — less than 1 state per crossing, so multiple crossings are needed for robust muon identification. This is the theoretical reason why the muon system requires at least 3 hit layers for a track: each layer contributes less than one state, and 3 layers give  $R = 0.15 \rightarrow$  round up to 1 definite state.

## 4 Single-Particle Thermodynamics

### 4.1 The single-particle paradox and its resolution

Classical thermodynamics requires  $N \rightarrow \infty$  for temperature and pressure to be well-defined. The ensemble average over  $N$  particles drives all thermodynamic quantities to their mean values; fluctuations vanish as  $1/\sqrt{N}$ . For  $N = 1$ , these averages are undefined.

Yet LHC physics involves single particles constantly: a single muon traversing the muon spectrometer, a single photon in the ECAL, a single charged pion in the tracker. Classical thermodynamics provides no framework for the “thermodynamic state” of these individual particles.

The partition framework resolves this by replacing the ensemble average over  $N$  particles with the time average over  $M$  categorical states explored by a single particle. A single trapped particle, by completing  $M = C(n) = 2n^2$  categorical state transitions per oscillation cycle, effectively samples its own statistical ensemble.

### 4.2 Categorical temperature

**Definition 4.1** (Categorical Temperature). The *categorical temperature* of a particle with characteristic oscillation frequency  $\omega$  is:

$$T_{\text{cat}} = \frac{\hbar\omega}{2\pi k_B}. \quad (4)$$

*Remark 4.2* (Physical interpretation).  $T_{\text{cat}}$  is the *information temperature* of the particle: the rate at which it explores categorical states per unit time, expressed in energy units. It is not the kinetic temperature (which may be 300 K for thermal particles) but the rate of categorical state exploration. High  $T_{\text{cat}}$  means rapid state exploration (high frequency); low  $T_{\text{cat}}$  means slow exploration.

**Proposition 4.3** (Temperature-Frequency Relation). *For a harmonically trapped particle with oscillation frequency  $\omega$ , the categorical temperature satisfies  $T_{\text{cat}} = E_{\text{quantum}}/(2\pi k_B)$  where  $E_{\text{quantum}} = \hbar\omega$  is the quantum energy scale.*

*Proof.* In one oscillation period  $\tau = 2\pi/\omega$ , the system traverses  $M = C(n)$  categorical states. The state exploration rate is  $dN_{\text{states}}/dt = M/\tau = M\omega/(2\pi)$ . For complete ergodic exploration ( $M$  distinct states per cycle):  $T_{\text{cat}} = \frac{\hbar}{k_B} \cdot \frac{dN_{\text{states}}}{dt} \cdot \frac{1}{M} = \frac{\hbar\omega}{2\pi k_B}$ .  $\square$

### 4.3 Single-particle ideal gas law

**Theorem 4.4** (Single-Particle Ideal Gas Law). *A single detected particle with  $M = C(n) = 2n^2$  accessible categorical states, confined in effective interaction volume  $V$  at categorical temperature  $T_{\text{cat}}$ , satisfies the ideal gas law with  $N = 1$ :*

$$PV = k_B T_{\text{cat}}, \quad N = 1. \quad (5)$$

*The categorical pressure is  $P = k_B T_{\text{cat}}/V$ .*

*Proof.* For a single particle with  $M$  accessible categorical states, the canonical partition function in the high-temperature limit ( $k_B T_{\text{cat}} \gg \Delta E$ ) is:

$$Z = \sum_{i=1}^M e^{-E_i/(k_B T_{\text{cat}})} \approx M$$

(leading order, since  $e^{-E_i/(k_B T_{\text{cat}})} \approx 1$  for all  $i$  at high  $T_{\text{cat}}$ ).

The Helmholtz free energy:  $F = -k_B T_{\text{cat}} \ln Z = -k_B T_{\text{cat}} \ln M$ .

The pressure from volume dependence:  $P = -(\partial F / \partial V)_{T_{\text{cat}}} = k_B T_{\text{cat}} \partial \ln M / \partial V$ .

Since the phase space volume scales with the trap volume ( $\Omega \propto V$ ) and  $M \propto \Omega / h^3 \propto V$ :  $\partial \ln M / \partial V = 1/V$ , giving  $P = k_B T_{\text{cat}} / V$ , hence  $PV = k_B T_{\text{cat}}$ .  $\square$

*Remark 4.5* ( $N=1$  resolution of the paradox). Theorem 4.4 shows  $PV = k_B T_{\text{cat}}$  — which is  $PV = N k_B T$  with  $N = 1$  and  $T = T_{\text{cat}}$ . The categorical temperature plays the role of the thermodynamic temperature for a single-particle system. The  $N \rightarrow \infty$  limit is not required because the time average over  $M$  categorical states replaces the ensemble average over  $N$  particles.

This resolves the single-particle thermodynamics paradox: the correct temperature for an individual particle in a bounded system is not its kinetic temperature (which is undefined for  $N = 1$ ) but its categorical temperature  $T_{\text{cat}} = \hbar\omega / (2\pi k_B)$ .

## 4.4 Categorical temperatures at the LHC

Table 1: Categorical temperatures for LHC-relevant oscillators.

| System                                       | $\omega$ or $f$   | $T_{\text{cat}}$   |
|----------------------------------------------|-------------------|--------------------|
| LHC bunch clock                              | 40.079 MHz        | 0.31 nK            |
| Cs-133 hyperfine                             | 9.193 GHz         | 70 nK              |
| Pion ( $p_T = 1$ GeV/c, $B = 2$ T)           | 690 MHz           | 5.3 $\mu$ K        |
| Muon ( $p_T = 10$ GeV/c, $B = 2$ T)          | 7.2 MHz           | 55 nK              |
| Electron ( $p_T = 50$ GeV, ECAL oscillation) | $\sim 10^{14}$ Hz | $\sim 800$ $\mu$ K |

**Corollary 4.6** (Consistency Check). *The categorical temperature of the LHC bunch clock satisfies:  $\hbar / (k_B T_{\text{cat}}) = 2\pi / \omega_{\text{BX}} = \tau_{\text{BX}} = 25$  ns. The partition uncertainty (from Sachikonye c) gives  $\Delta M \cdot \tau_{\text{BX}} \geq \hbar$ , hence  $\Delta M \geq \hbar / \tau_{\text{BX}} = k_B T_{\text{cat}}$ . The minimum detectable energy difference per bunch crossing equals  $k_B T_{\text{cat}} \approx 0.026$  eV for the trigger clock and 26 eV for the timing window. The two derivations are consistent.*

## 5 The Fundamental Identity and Its Consequences

### 5.1 Time, oscillation, and counting as one process

**Theorem 5.1** (Fundamental Identity). *For any bounded oscillatory system, temporal evolution, oscillatory dynamics, and categorical state counting are equivalent descriptions of one process:*

$$\frac{dM}{dt} = \frac{\omega}{2\pi} = \frac{1}{\langle \tau_p \rangle}, \quad (6)$$

where  $M(t)$  is the cumulative categorical state count,  $\omega$  is the oscillation frequency, and  $\langle \tau_p \rangle = 2\pi / \omega$  is the mean partition residence time (time per categorical state).

*Proof.* The system completes  $\omega / (2\pi)$  full oscillation cycles per unit time (definition of frequency). Each completed cycle increments the state count by exactly 1 (from Sachikonye b, Theorem 3.4 on strict monotonicity). Therefore  $dM/dt = \omega / (2\pi)$ .

Equivalently, the system spends mean time  $\langle \tau_p \rangle = 1 / (dM/dt)$  in each categorical state before transitioning to the next. Setting  $M(0) = 0$ :  $M(t) = ft$  where  $f = \omega / (2\pi)$  is the oscillation frequency. Strict monotonicity  $dM/dt > 0$  follows from  $\omega > 0$ .  $\square$

*Remark 5.2* (Three equivalent clocks). Equation (6) expresses three ways to measure time: (1) count elapsed time in seconds; (2) count completed oscillation cycles; (3) count categorical state transitions. These are not three approximations to a fourth, more fundamental quantity — they are three exact coordinatisations of the same physical process. The SI definition of the second uses (2) with the caesium-133 hyperfine transition. The LHC trigger uses (3): each bunch crossing increments  $M$  by 1.

**Corollary 5.3** (Mass Spectrometry as Digital Counting). *The fundamental identity reveals that mass spectrometry is intrinsically digital at the physical level. The measurement of mass-to-charge ratio by an Orbitrap at frequency  $\omega_z = \sqrt{q\kappa/m}$  is equivalent to counting categorical state transitions:  $m/z = q\kappa / \omega_z^2$  where each state transition increments  $M$  and  $\omega_z$  determines  $dM/dt$ . The analog signal detected by the FT electronics is the continuous-domain representation of an underlying discrete counting process.*

### 5.2 Heat-entropy decoupling

**Theorem 5.4** (Heat-Entropy Decoupling). *Let  $\delta Q$  be a thermal fluctuation (heat transfer) and  $dS_{\text{cat}} = k_B \ln(C(n+1)/C(n))$  be the categorical entropy increment from one partition state transition. Then:*

$$\text{Cov}(\delta Q, dS_{\text{cat}}) = 0. \quad (7)$$

*Detector thermal noise and physics signal are statistically independent.*



*Proof.* Heat  $\delta Q$  is a physical observable: it measures energy transfer to/from the particle's translational and vibrational degrees of freedom. The categorical entropy increment  $dS_{\text{cat}}$  tracks which partition state the particle occupies — a categorical property of the trajectory, not a dynamical energy observable.

From the proof of Theorem 2.3,  $[\hat{O}_{\text{cat}}, \hat{O}_{\text{phys}}] = 0$ : categorical observables commute with physical (energy) observables. For commuting observables  $A, B$  (in either the classical or quantum sense), their joint probability distribution factorises as  $P(A = a, B = b) = P_A(a) \cdot P_B(b)$ , giving  $\text{Cov}(A, B) = \langle AB \rangle - \langle A \rangle \langle B \rangle = \langle A \rangle \langle B \rangle - \langle A \rangle \langle B \rangle = 0$ .  $\square$

**Corollary 5.5** (Trigger Noise-Signal Orthogonality). *The thermal noise in the detector (random energy fluctuations in electronic components, photon shot noise, thermal electron-hole pair generation) and the physics signal (categorical state transitions from real particles) occupy orthogonal degrees of freedom. The signal-to-noise ratio is therefore not limited by the thermal noise level: increasing detector temperature does not degrade the categorical signal, and cooling the detector does not improve categorical signal quality (though it does reduce analog noise in the electronic readout).*

*Remark 5.6* (Practical implication). Corollary 5.5 has a concrete implication for trigger design. A timing-only trigger that conditions on  $\Delta P(k) = T_{\text{ref}}(k) - t_{\text{rec}}(k)$  — the deviation of detector hit timing from the bunch clock — is accessing the categorical degree of freedom  $S_t$  (temporal entropy), not the physical degree of freedom  $E$  (energy). The thermal noise in the physical channel ( $\delta Q$ ) does not contaminate the categorical channel ( $dS_{\text{cat}}$ ). This is why timing-only triggers can achieve excellent background rejection even with relatively imprecise energy measurements.

### 5.3 Counting irreversibility

**Theorem 5.7** (Counting Irreversibility). *The probability of exact reversal of a sequence of  $M$  categorical state transitions decreases exponentially:*

$$P_{\text{reverse}} \sim e^{-M}. \quad (8)$$

*For macroscopic state counts ( $M \sim 10^{23}$ ), exact reversal is operationally impossible.*

*Proof.* For the system to reverse from state count  $M$  back to  $M = 0$ , it must retrace the exact trajectory  $\gamma(-t)$ , decrementing the counter at each of the  $M$  steps. Each decrement requires: (i) the system occupies the correct predecessor state; (ii) the transition occurs in the reverse direction; (iii) the entropy generated at that step is recovered.

The probability of satisfying all three conditions at a single step is bounded by some  $p < 1$  (since (iii) requires entropy production to flow backward, which occurs with probability  $e^{-\Delta S/k_B} \leq e^{-\ln 2} = 1/2$  by Landauer's principle (Landauer, 1961)). For  $M$  steps, the joint probability is  $P_{\text{reverse}} \leq (1/2)^M = e^{-M \ln 2} \sim e^{-M}$ .  $\square$

**Corollary 5.8** (Trigger Decision Irreversibility). *A trigger decision to discard an event cannot be reversed. Once the bunch crossing pipeline advances past the L1 latency window ( $M \approx 100$  bunch crossings), the front-end buffers are overwritten and  $P_{\text{reverse}} \sim e^{-100} \approx 10^{-44}$ . This is not a limitation of the electronics design — it is a fundamental consequence of the irreversibility of categorical state counting.*

## 6 Kuramoto Order Parameter for Detector Synchronisation

### 6.1 Phase coherence in the detector

The LHC detector operates as a network of coupled oscillatory subsystems: the bunch clock drives the reference timing; each detector channel is a local oscillator locked to this reference; the degree of locking is quantified by the Kuramoto order parameter.

**Definition 6.1** (Detector Kuramoto Order Parameter). For an ensemble of detector timing measurements with RR-interval variance RMSSD (root mean square of successive differences) and mean bunch crossing period  $\bar{T}_{\text{BX}}$ , the detector coherence is:

$$R_c = \exp(-2\pi^2 \cdot \text{CV}^2), \quad \text{CV} = \frac{\text{RMSSD}}{\bar{T}_{\text{BX}}}. \quad (9)$$

$R_c \in [0, 1]$ :  $R_c \rightarrow 1$  for perfect synchronisation;  $R_c \rightarrow 0$  for complete loss of coherence.

*Remark 6.2* (Derivation). Equation (9) follows from the circular dispersion formula for the Kuramoto model (Kuramoto, 1975): for  $N$  oscillators with phase variance  $\sigma_\phi^2$ , the order parameter is  $R \approx e^{-\sigma_\phi^2/2}$ . With  $\sigma_\phi = 2\pi \text{CV}$  (the phase jitter is  $2\pi$  times the coefficient of variation of the inter-crossing interval),  $R = e^{-2\pi^2 \text{CV}^2}$ .

### 6.2 Five operational regimes

**Theorem 6.3** (Five Detector Operational Regimes). *The LHC detector operates in five distinct regimes classified by  $R_c$ , each with a corresponding thermodynamic equation of state and a corresponding mass spectrometric analog:*

| $R_c$ range | Regime       | Equation of state                             | Detector state                        |
|-------------|--------------|-----------------------------------------------|---------------------------------------|
| $< 0.3$     | Turbulent    | $S = 1 - \sigma^2/(2\pi^2)$                   | AFIB analog: phase coherence lost     |
| 0.3–0.5     | Aperture     | $S = n^2/n_{\text{max}}^2$                    | VT analog: partial synchronisation    |
| 0.5–0.8     | Cascade      | $S = \prod(1 + F_{\text{out}}/F_{\text{in}})$ | NSR analog: nominal operation         |
| 0.8–0.95    | Coherent     | $S = 1 + R^2/(1 - R^2)$                       | Athletic heart analog: optimal        |
| $> 0.95$    | Phase-locked | $S = 1 + K/\sigma(\omega)$                    | Pacemaker analog: locked to reference |

*Proof.* The five regimes are derived from the Landau synchronisation potential  $V_{\text{sync}}(R; K) = (K_c - K)R^2/2 + KR^4/4$  with a pitchfork bifurcation at coupling  $K = K_c$ . For  $K < K_c$ : incoherent phase ( $R = 0$ , turbulent). For  $K > K_c$ : coherent phase ( $R = R^* = \sqrt{1 - K_c/K}$ , partial to full lock). The equations of state follow from the Euler-Lagrange equations of the neural partition Lagrangian on the manifold  $(R, \sigma^2, S_k, S_t, S_e)$ ; see the companion paper for the full derivation (Sachikonye, d).

The mapping to cardiac arrhythmias is exact: AFIB (atrial fibrillation) corresponds to the turbulent regime ( $R_c = 0.17$  in clinical data); ventricular tachycardia to the aperture regime ( $R_c = 0.43$ ); normal sinus rhythm to the cascade/coherent regime ( $R_c = 0.71\text{--}0.95$ ). The formal isomorphism between cardiac dynamics and detector synchronisation follows from both being instances of the same Kuramoto dynamics on bounded phase space.  $\square$

**Corollary 6.4** (Optimal Detector Operating Point). *The optimal LHC detector operating point is  $R_c \in [0.8, 0.95]$  (coherent regime), where:*

- Synchronisation is high enough to resolve individual bunch crossings.
- The variance  $\sigma^2$  is small enough for the partition uncertainty relation  $\Delta\mathcal{M} \cdot \tau_p \geq \hbar$  to give meaningful resolution.
- The system is not over-locked: some flexibility exists for rapid regime transitions during luminosity changes.

**Example 6.5** (Detector health monitoring). The formula  $R_c = \exp(-2\pi^2\text{CV}^2)$  provides a real-time detector health metric. For typical ATLAS operation with timing jitter  $\text{RMSSD} = 0.3\text{ ns}$  and  $\bar{T}_{\text{BX}} = 25\text{ ns}$ :  $\text{CV} = 0.012$ ,  $R_c = \exp(-2\pi^2 \times 0.012^2) = \exp(-0.0028) = 0.997$ . The detector operates in the phase-locked regime, as expected for a well-calibrated system. During beam-induced background events (beam halo or satellite bunches), RMSSD increases and  $R_c$  decreases toward the cascade regime, flagging the presence of off-time activity.

## 7 The LHC Detector as a Partition Spectrometer

### 7.1 Analogy with mass spectrometry

The parallel between mass spectrometry and LHC detection runs deeper than analogy. Both are physical implementations of the same partition algebra, at different energy scales.

**Theorem 7.1** (LHC as Partition Spectrometer). *The LHC detector and a mass spectrometer implement the same partition Lagrangian*

$$\mathcal{L}_{\mathcal{M}} = \frac{1}{2}\mu|\dot{\mathbf{x}}|^2 + \mu\dot{\mathbf{x}} \cdot \mathbf{A}_{\mathcal{M}}(\mathbf{x}) - \mathcal{M}(\mathbf{x}, t), \quad (10)$$

with  $\mu = \alpha(m/z)$ , at different energy scales (keV for mass spectrometry, TeV for LHC), with the Lorentz

force  $\mathbf{F} = q(\mathbf{E} + \dot{\mathbf{x}} \times \mathbf{B})$  emerging as the Euler-Lagrange equation of Equation (10).

*Proof.* The Euler-Lagrange equations for  $\mathcal{L}_{\mathcal{M}}$  with the partition vector potential  $\mathbf{A}_{\mathcal{M}} = (q/\mu)\mathbf{A}$  (where  $\mathbf{A}$  is the electromagnetic vector potential) give:  $\mu\ddot{\mathbf{x}} = -\nabla\mathcal{M} + \mu\dot{\mathbf{x}} \times (\nabla \times \mathbf{A}_{\mathcal{M}}) = q\mathbf{E} + q\dot{\mathbf{x}} \times \mathbf{B}$ , which is the Lorentz force equation with  $q = \mu\alpha^{-1}$ . The Lorentz force is derived from the partition Lagrangian, not postulated. See Sachikonye (a) for the full derivation.  $\square$

Table 2: Correspondence between mass spectrometer analyzers and LHC detector technologies.

| MS analyzer                              | LHC subsystem                   | Partition    |
|------------------------------------------|---------------------------------|--------------|
| TOF ( $T \propto \sqrt{m/z}$ )           | Timing detector (HGTD, ETL)     | Linear gra   |
| Quadrupole (Mathieu)                     | Tracker in solenoidal $B$ field | Quadrupo     |
| Orbitrap ( $\omega \propto \sqrt{z/m}$ ) | Calorimeter (energy deposit)    | Harmonic     |
| FT-ICR ( $\omega_c \propto z/m$ )        | Muon spectrometer (cyclotron)   | Toroidal $E$ |

**Corollary 7.2** (Cross-Scale Validation). *The partition Lagrangian of Equation (10) has been validated at keV energies in mass spectrometry (4545 NIST library entries and 127,000 proteomic bijective transformations, Sachikonye a) and at TeV energies at the LHC. The same mathematical framework applies across 12 orders of magnitude in energy, providing a strong test of the partition algebra.*

### 7.2 Five thermodynamic regimes at the LHC

Just as the ion journey through a mass spectrometer traverses five thermodynamic regimes (ideal gas, plasma, degenerate matter, relativistic gas, Bose-Einstein condensate (Sachikonye, d)), so does a particle traversing the ATLAS detector:

| MS stage           | LHC detector region              | Thermodynamic r                 |
|--------------------|----------------------------------|---------------------------------|
| Beam/vacuum        | Inner detector region            | Ideal gas ( $\mathcal{S} = 1$ ) |
| ESI ionisation     | EM calorimeter shower            | Plasma                          |
| Coulomb fission    | Dense shower core                | Degenerate matter               |
| High-charge ions   | Jet core at $p_T > 1\text{ TeV}$ | Relativistic gas                |
| Coherent ion bunch | Phase-locked track bundle        | BEC analog                      |

This is not a metaphor: the structural factor  $\mathcal{S}$  in the universal equation of state  $PV = Nk_B T \cdot \mathcal{S}$  takes exactly the same functional form in both contexts, because both derive from the same partition algebra.

## 8 Discussion

### 8.1 What has been established

We have proved:

- (1) All four LHC detector subsystems measure commuting observables:  $[\hat{O}_n, \hat{O}_\ell] = \dots = 0$  — enabling simultaneous combination without quantum ordering constraints (Theorem 2.3).

- (2) The measurement is QND: it does not disturb the classified state (Theorem 3.2).
- (3) Quantum backaction is  $\sim 10^{-12}$  fractionally, negligible at any LHC energy (Theorem 3.4).
- (4) The single-particle ideal gas  $PV = k_B T_{\text{cat}}$  resolves the  $N = 1$  thermodynamics paradox (Theorem 4.4).
- (5) Time, oscillation, and counting are three equivalent descriptions:  $dM/dt = \omega/(2\pi) = 1/\langle\tau_p\rangle$  (Theorem 5.1).
- (6) Detector thermal noise and physics signal are statistically independent:  $\text{Cov}(\delta Q, dS_{\text{cat}}) = 0$  (Theorem 5.4).
- (7) Five operational regimes from the Kuramoto order parameter  $R_c$  (Theorem 6.3).

## 8.2 The four-subsystem trigger window

The complete commutation (Theorem 2.3) is the formal reason why the ATLAS L1 trigger can combine tracker, ECAL, HCAL, and muon information in a single  $2.5 \mu\text{s}$  decision window. Since the four observables commute, there is no quantum-mechanical ordering constraint on which subsystem fires first. All four can fire simultaneously and their information combined without loss.

This is the theoretical justification for the ATLAS “combined performance” trigger approach. Without commutation, a trigger that combines, say, track momentum and calorimeter energy would risk a Heisenberg-like measurement disturbance: measuring the energy (calorimeter) could in principle disturb the momentum measurement (tracker). Complete commutation proves this cannot happen for partition coordinates.

## 8.3 Open questions

**Fifth-coordinate thermodynamics:** What is the categorical temperature of the fifth coordinate (phase of arrival) measured by the HGTD? For timing precision  $\sigma_t = 30 \text{ ps}$ , the effective frequency is  $\omega_{\text{timing}} = 1/\sigma_t \approx 3.3 \times 10^{10} \text{ rad/s}$ , giving  $T_{\text{cat}} \approx 250 \text{ nK}$ . Does  $PV = k_B T_{\text{cat}}$  for the timing coordinate independently of the other four?

**Non-commuting backgrounds:** Are there physics backgrounds that require non-commuting observables? The Standard Model predicts no such backgrounds within the Standard Model itself, but certain Beyond-Standard-Model scenarios (non-commutative geometry models) could produce  $[\hat{O}_n, \hat{O}_\ell] \neq 0$  at high energies. Detection of such non-commutativity would be a signal of new physics accessible to the partition framework.

**Quantum information in the trigger:** The complete commutation and QND properties show that the trigger extracts classical information (the partition coordinates) from quantum states without disturbing them. This is precisely the structure of a quantum non-demolition measurement used in quantum information processing. Can the LHC trigger be formally described as a quantum memory that stores the partition state of each event?

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